





Course Introduction

Course Organization

Week 1

- Big data foundations
- Course package introduction
- Python introduction

Week 2

- Supervised vs Unsupervised learning
- Mathematical foundations of machine learning

Week 3

- Curse of dimensionality
- Why having multiple variables is not always good?
- **Group assignment 1**

Course Organization

Week 4

- Principal component analysis
- Singular Value Decomposition

Week 5

- Embeddings
- t-SNE
- **Group assignment 2**

Week 6

- Clustering algorithms
- KMeans

Course Organization

Week 7

- DBScan and hierarchical clustering
- **Midterm assignment**

Week 8

- **Midterm exam**

Week 9

- Recommendation systems
- Content based recommendations

Course Organization

Week 10

- Collaborative filtering

Week 11

- Hybrid recommendations
- Other recommendation algorithms
- **Group assignment 3**

Week 12

- Graph analytics foundations

Course Organization

Week 13

- Page rank algorithm
- **Group assignment 3**

Week 14

- Machine learning pipelines

Week 15

- Deployment of algorithms
- **Final group assignment**



Prof Introduction



Students introduction



Character introduction



Yunguri, the llama

She likes music,
statistics and data
analytics



Yunguri, the peccary

He likes music, coding
and data analytics



called a problem
child cuz I got too wild,
but now that's how I'm
getting paid





called a problem
child cuz I got too wild,
but now that's how I'm
getting paid



I'm done hiding, now
I am shining, like a born
to beeeeeeee





Hey Mathias, that song
is really nice, it really
cheer me up





Hey Mathias, that song
is really nice, it really
cheer me up



I like it too! I love EJAЕ.
Actually, I really need
something to cheer me
up





What happened
Mathias? Is everything
ok?





What happened Mathias? Is everything ok?



Of course not, it's time to go back to classes, and summer is finishing





I have a genius idea. A
playlist assistant for
students based on the
mood





And we can use data to
make it really
interesting





Ok, I like the idea, but
what data do we have?
what data do we wish
we had? do we need
big data? what's big
data?



And we can use data to
make it really
interesting





What's big data?

Big data is like teenage sex:
everyone talks about it, nobody
really knows how to do it,
everyone thinks everyone else is
doing it, so everyone claims they
are doing it.

Dan Ariely



Seriously, what's big data?

Big data is not hype. For something to be considered big data, it has to fulfill the usual five Vs.

V	Description
Volume	Is the data big enough?
Velocity	Is the data being generated in a fast pace?
Variety	Is the data diverse?
Veracity	Is the data reliable?
Value	Is the data meaningful?



Seriously, what’s big data?

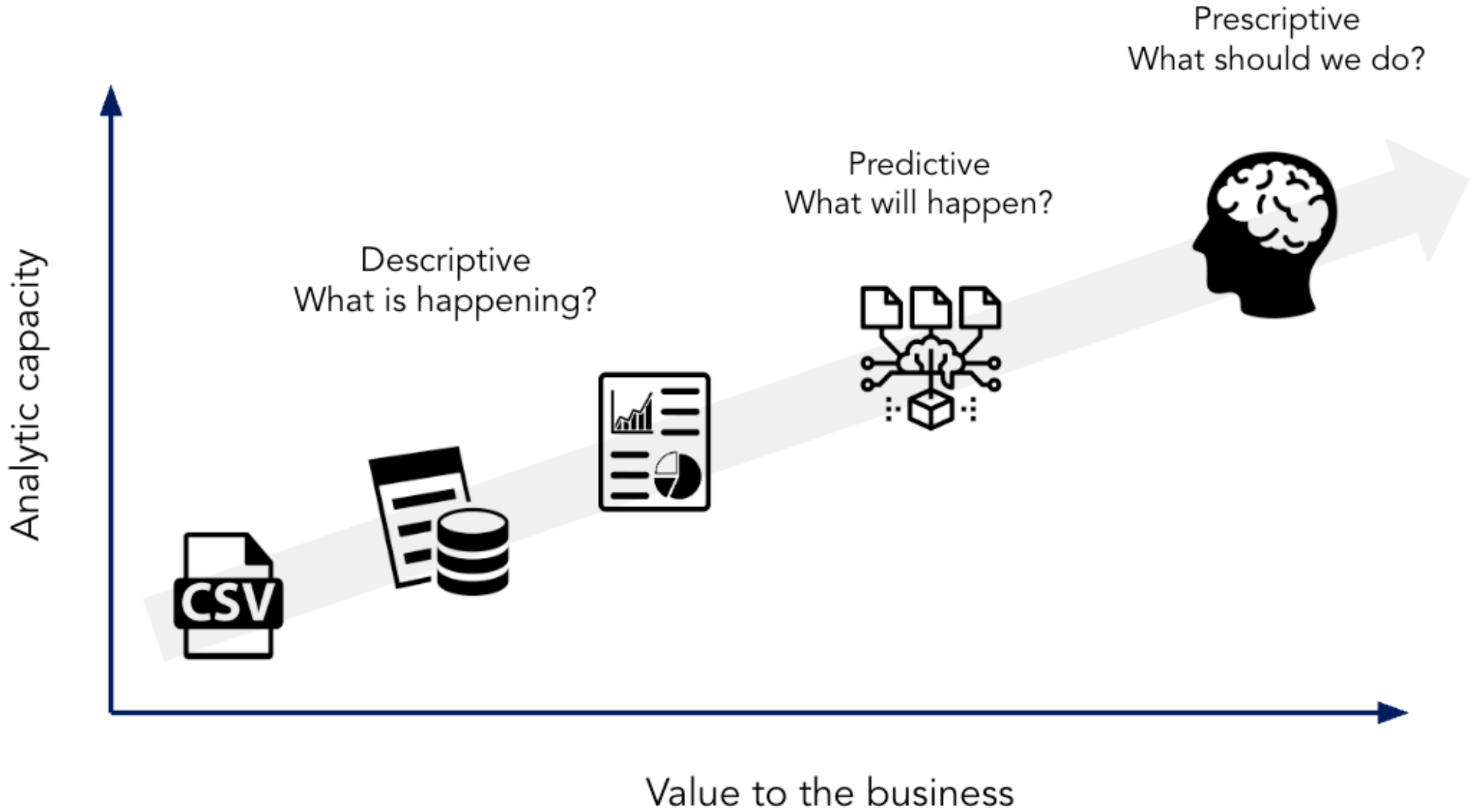
A dataset is not big because it looks intimidating. It is “big” when at least one of the following changes:

Element	Description
Computational architecture	Do we need a different computational architecture?
Storage strategy	Do we need to save the data in a different way?
The algorithmic design	Do we need a different strategy to train the model?
The evaluation protocol	The criteria of success is different because of the size of the data?



Then, what's data analytics?

There are different levels of analytics





The data we will use



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upc-datasets 0.6.0

`pip install upc-datasets`

Released: about 17 hours ago

Student-facing structured dataset toolkit for the UPC big data course.

Navigation

Project description

Release history

Download files

Verified details 

These details have been [verified by PyPI](#)

Maintainers

 [aladelca16](#)

Unverified details

These details have **not** been verified by PyPI

Project links

Project description

UPC Datasets

This workspace contains a small Python toolkit for generating structured teaching datasets for the `PachaMix` course narrative.

The implementation is designed around:

- structured tables
- metadata
- audio-feature tables
- lyrics-derived features
- playlist interactions
- parquet outputs

It intentionally avoids:

- raw mp3 processing
- waveform pipelines

 ADRIAN ALARCON · UPDATED 15 HOURS AGO

0

<> Code

Download



Data Card

Code (0)

Discussion (0)

Suggestions (0)

Settings

Pending Actions

USABILITY SCORE: 4.71

Add tags

Make it easy for users to find your dataset in search

Upload an image

Make your dataset pop with an eye-catching cover image and thumbnail

Add file information

Help others navigate your dataset with a description of each file

Include column descriptors

Empowers others to understand your data by describing its features

About Dataset

Edit

Usability 

4.71

License

Other (specified in descripti... [Edit](#)

Expected update frequency

Not specified [Edit](#)

Tags

Add Tags

PachaMix Playlist Datasets

This dataset publishes the playlist-behavior tables used by the `PachaMix` teaching toolkit for recommendation, popularity analysis, and graph construction.

It is the `Kaggle` distribution channel for the large playlist-derived artifacts, while smaller course datasets are distributed separately through package/release assets.

What is included

`playlist_events.parquet`

- Shape: 125,451,006 rows x 8 columns
- File size: about 4.5 GB
- Grain: one row per (playlist_id, track_uri, position)

This is the main interaction table. Each row says that a track appears in a playlist at a given position.

playlist_stats.parquet

- Shape: 1,050,816 rows x 3 columns



Tools that we will use



CODE

SLEEP

REPEAT